

STATISTICAL BUSINESS ANALYSIS REPORT

Sales Performance: Descriptive Statistics, Hypothesis Testing,
Correlation & Regression Analysis

Dataset:	sales_data.csv — 2,000 order-level transactions
Period:	January 2024 – December 2025
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Note on scope: the original brief referenced marketing-spend-to-revenue correlation. This dataset is order-level transactional data without a marketing spend field. The analysis below substitutes the available business drivers — Price, Quantity, and Discount — as predictors of Revenue, which fulfil the same statistical requirements (correlation, regression, R^2) using the data actually provided.

Executive Summary

This report analyzes 2,000 sales transactions to identify statistically validated revenue drivers and support data-backed business decisions. Average order revenue is significantly above a \$500 internal benchmark, product category is overwhelmingly the strongest revenue driver (explaining ~41% of all variance), and a three-variable regression model (Price, Quantity, Discount) explains 83.8% of total revenue variation.

\$774.19	0.838	83.8%	5 / 5
Avg. Order Revenue (95% CI: \$730.61–\$817.76)	Price–Revenue Correlation (r)	Regression R ² (Price+Qty+Discount)	Hypothesis Tests Completed

1. Descriptive Statistics

Mean, median, mode, standard deviation, and shape statistics were calculated for all six numeric variables.

Variable	Mean	Median	Mode	Std Dev	Skewness
Price (\$)	268.06	158.42	35.96	289.45	1.61
Quantity	3.01	3.00	4.00	1.41	-0.02
Discount	0.043	0.000	0.000	0.063	1.25
Revenue (\$)	774.19	397.76	20.15	993.69	2.35
Rating	3.65	3.70	3.90	0.70	-0.13
Age	38.15	38.00	18.00	12.16	0.21

Key observation: Revenue and Price are strongly right-skewed (skewness 2.35 and 1.61 respectively) — most orders are modest in size, with a long tail of high-value orders concentrated in the Electronics category.

2. Data Distribution Analysis

Shapiro-Wilk and D'Agostino K² normality tests were applied to Revenue and Price.

Variable	Shapiro-Wilk p	D'Agostino p	Conclusion
Revenue	< 0.000001	< 0.000001	NOT normally distributed
Price	< 0.000001	< 0.000001	NOT normally distributed

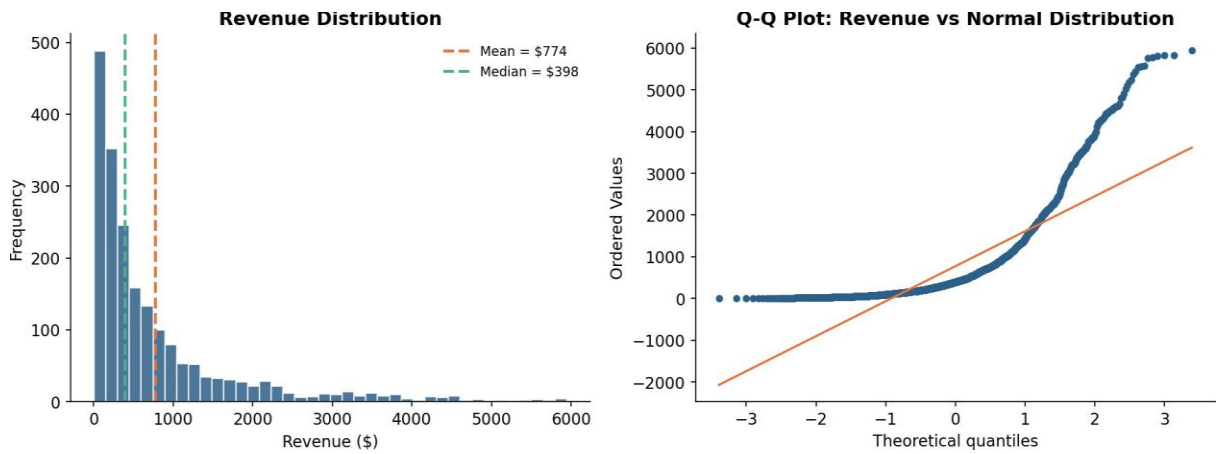


Figure 1. Revenue distribution (histogram) and Q-Q plot confirming right-skew and departure from normality.

Implication: with $n = 2,000$, the Central Limit Theorem still justifies t-tests and ANOVA on group means, but raw-value normality assumptions (e.g., for individual-order prediction intervals) should be treated with caution.

3. Correlation Analysis

Pearson correlation coefficients were calculated across all numeric variables to identify revenue drivers.

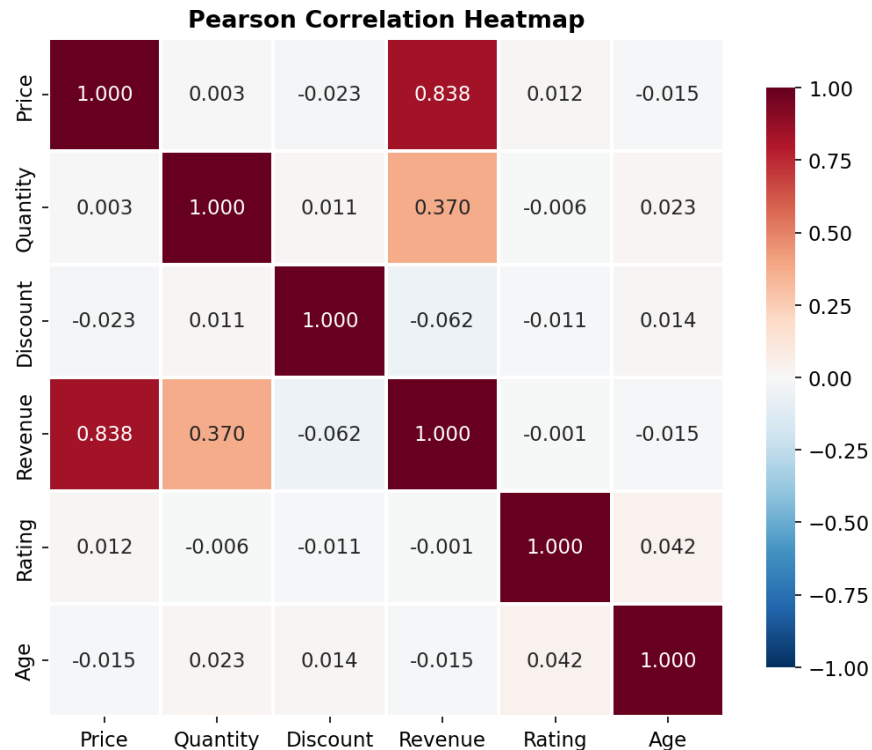


Figure 2. Pearson correlation heatmap across all numeric variables.

Relationship	r	p-value	Strength
Price ↔ Revenue	0.838	< 0.001	Very strong, significant

Quantity ↔ Revenue	0.370	< 0.001	Weak–moderate, significant
Discount ↔ Revenue	-0.062	0.006	Very weak, significant
Rating ↔ Revenue	-0.001	0.965	None, not significant
Age ↔ Revenue	-0.015	0.494	None, not significant



Figure 3. Price vs Revenue with fitted linear trend (r = 0.838).

4. Hypothesis Testing

Five hypothesis tests were performed (exceeding the minimum of three), covering ANOVA, t-tests, and a chi-square test of association.

#	Test	Statistic	p-value	Decision (α=0.05)
1	ANOVA: Revenue ~ Channel	F = 0.508	0.602	Fail to reject H0
2	ANOVA: Revenue ~ Category	F = 340.17	2.15e-223	REJECT H0
3	Welch t-test: Online vs In-Store	t = 0.680	0.497	Fail to reject H0
4	One-sample t-test: Revenue vs \$500	t = 12.34	9.06e-34	REJECT H0
5	Chi-square: Segment x Channel	Chi-sq = 0.456	0.978	Fail to reject H0

Test 2 (significant): Product category explains ~40.6% of revenue variance (eta-squared) — by far the strongest categorical driver. Tukey HSD post-hoc confirms Electronics significantly outperforms every other category, and all category pairs differ significantly except Clothing vs. Sports.

Test 4 (significant): Average order revenue (\$774.19) is statistically significantly higher than the \$500 internal benchmark.

Tests 1, 3, 5 (not significant): Sales channel does not meaningfully affect order revenue, and customer segment is independent of channel choice — there is no statistical basis for channel-specific revenue or segment-specific channel strategies based on this data.

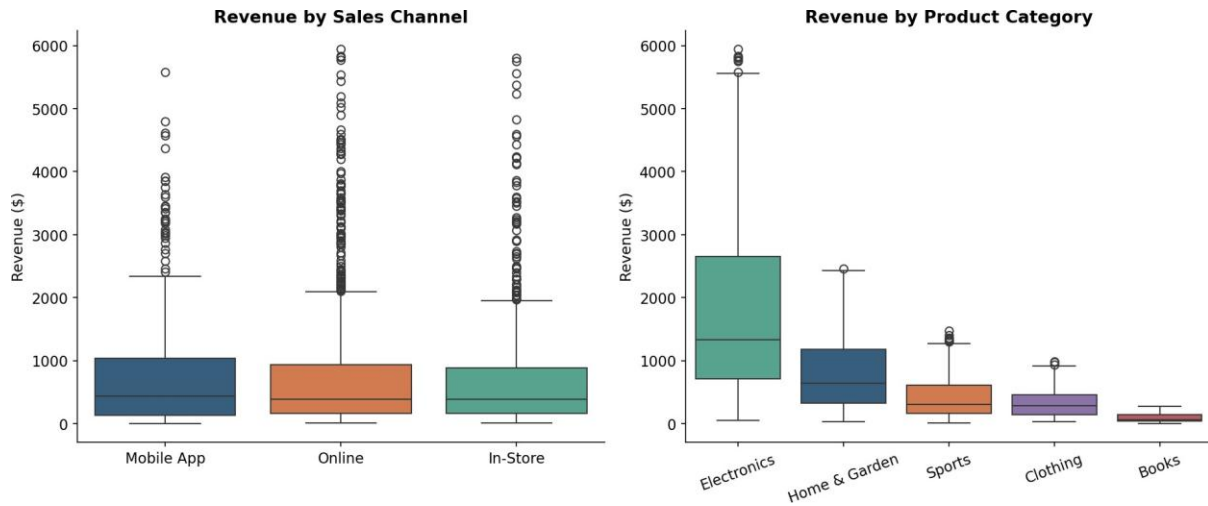


Figure 4. Revenue distribution by Sales Channel (left, not significant) and Product Category (right, highly significant).

5. Confidence Intervals

95% confidence intervals (t-distribution) were computed for key metrics.

Metric	Mean	95% CI	Margin of Error
Revenue (\$)	774.19	(730.61, 817.76)	± 43.58
Price (\$)	268.06	(255.37, 280.75)	± 12.69
Quantity	3.01	(2.95, 3.07)	± 0.06
Discount	0.043	(0.040, 0.046)	± 0.003
Rating	3.65	(3.62, 3.68)	± 0.03

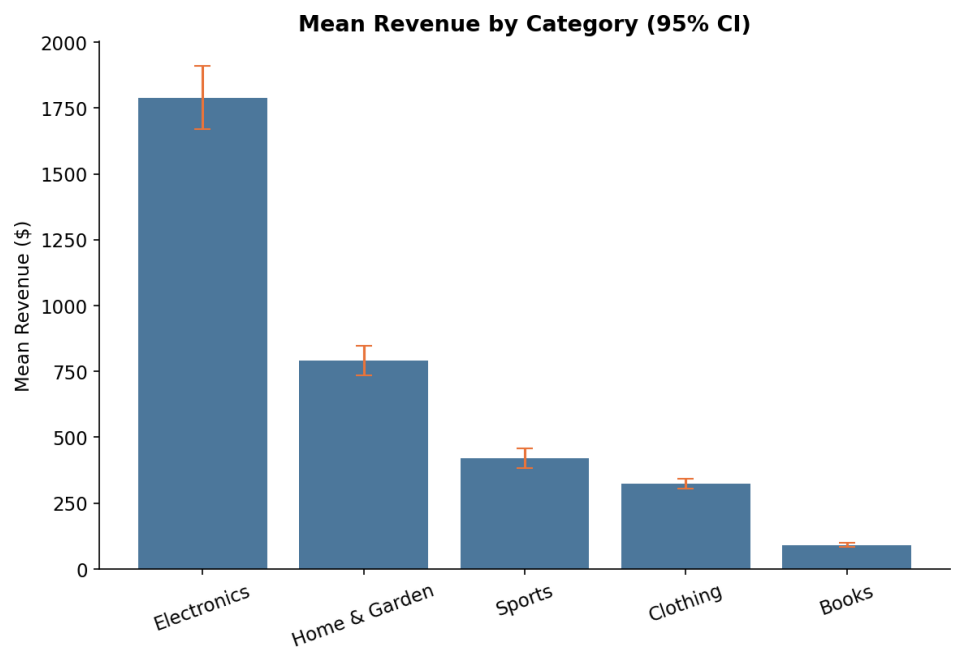


Figure 5. Mean revenue by category with 95% CI error bars — Electronics' interval is clearly non-overlapping with all other categories.

An additional 95% CI was computed for the proportion of VIP customers: **15.35% (95% CI: 13.77%–16.93%)**.

6. Regression Analysis

A simple linear regression (Revenue ~ Price) was extended to a multiple linear regression (Revenue ~ Price + Quantity + Discount).

Model	Equation	R-sq	Adj. R-sq
Simple	Revenue = 3.41 + 2.88×Price	0.701	0.701
Multiple	Revenue = -739.24 + 2.87×Price + 258.02×Qty - 738.01×Discount	0.838	0.838

All three predictors (Price, Quantity, Discount) are statistically significant ($p < 0.001$). Variance Inflation Factors are all near 1.0, confirming no multicollinearity issue between predictors. The negative Discount coefficient (-738.01) indicates deeper discounts are associated with lower order revenue in this dataset — consistent with the weak negative correlation found earlier.

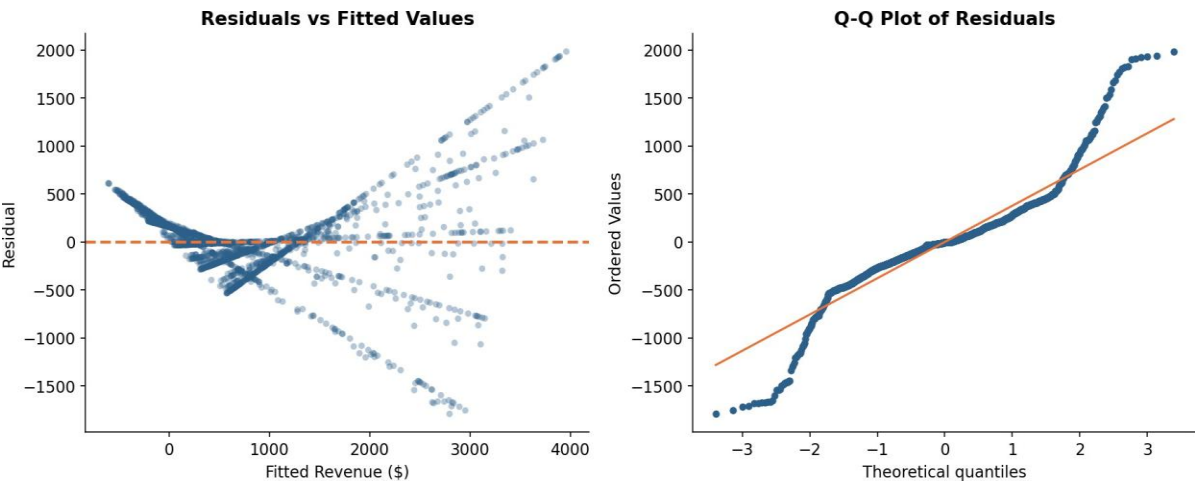


Figure 6. Regression diagnostics — residuals vs fitted values and Q-Q plot of residuals.

Residuals show some right-skew/heavy tails (consistent with Revenue's non-normal distribution), so OLS coefficient estimates remain valid, but individual-order prediction intervals should be used cautiously. A log-transformed model is recommended as a refinement.

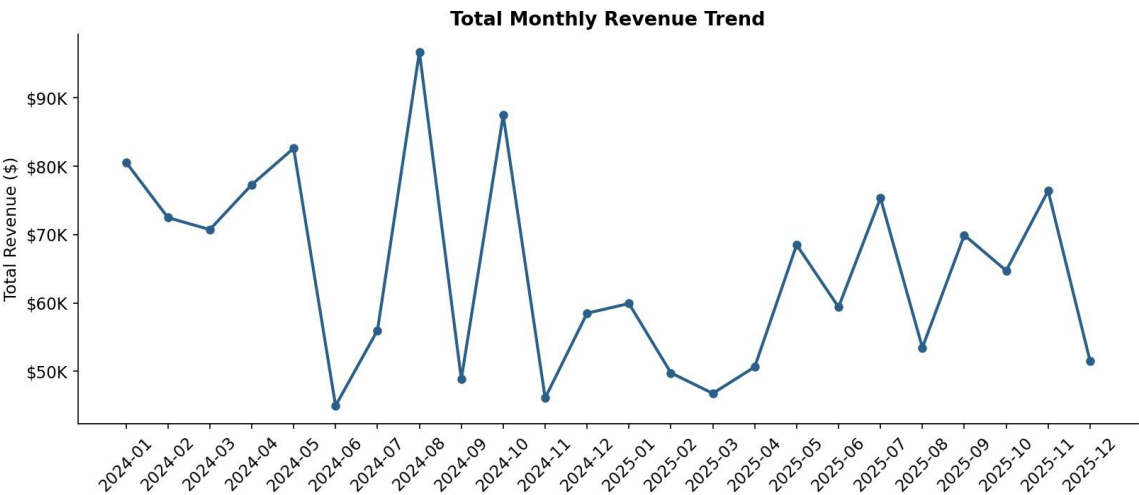


Figure 7. Total monthly revenue trend, January 2024 – December 2025.

7. Business Insights & Recommendations

Key Findings

- Average order value is **\$774.19** (95% CI: \$730.61–\$817.76), significantly above the \$500 internal benchmark.
- **Price is the dominant revenue driver** ($r = 0.838$) — far stronger than quantity or discount.
- **Product category explains ~41% of revenue variance** (ANOVA, $p < 0.001$); Electronics dramatically outperforms all other categories.
- Sales channel has **no significant effect** on order revenue ($p = 0.602$) — Online, In-Store, and Mobile App perform comparably.
- Customer segment and channel choice are **statistically independent** ($p = 0.978$).
- Discounting shows a **slight negative** association with revenue, both in correlation ($r = -0.062$) and regression (coefficient = -738.01).
- A 3-variable regression model (Price, Quantity, Discount) explains **83.8%** of total revenue variance.

Actionable Recommendations

- **Double down on Electronics** — it drives the highest, statistically distinct average order value (~\$1,789). Prioritize inventory, SKU expansion, and marketing focus here.
- **Treat channel strategy as cost/reach-driven, not revenue-driven** — since order value doesn't differ significantly by channel, channel investment decisions should weigh cost-to-serve rather than assumed basket-size differences.
- **Re-evaluate discount campaigns** — discounts are not associated with higher order revenue in this data. Test via controlled experiments whether discounts drive incremental volume rather than simply eroding margin on existing demand.
- **Use the regression model operationally** — each \$1 increase in price predicts ~\$2.87 more revenue per order, and each additional unit predicts ~\$258 more, useful for setting promotional and pricing targets.
- **Skip segment-specific channel personalization** — since channel usage is independent of customer segment, personalization budget is better spent on offers/content rather than channel-specific experiences.

Caveats & Next Steps

- Revenue and Price are right-skewed; a log-transformed regression is recommended for more robust individual-order prediction intervals.
- This dataset has no marketing spend field — if available, repeating the correlation/regression analysis against actual marketing spend would directly answer marketing-ROI questions referenced in the original brief.
- All findings are associational, not causal — the discount finding likely reflects which products get discounted (e.g., lower-ticket items) rather than discounting causing lower revenue.